MECH5200 Numerical Methods for PDEs : *Video 19b: Introduction to Finite Volume Methods – 2 Dimensions*

David Willis

March 6, 2026

References and Acknowledgements

The following materials were used in the preparation of this lecture:

1. Randall J. Leveque *Numerical Methods for Conservation Laws*.
2. Randall J. Leveque *Finite Volume Methods for Hyperbolic Problems*.
3. Morton and Meyers, *Numerical Solution of Partial Differential Equations*.
4. 16.920, lecture 11,12 Notes
5. Sethian, J.A., Level Set Methods – Evolving interfaces in Geometry, Fluid Mechanics
6. Versteeg and Malalasekera, *An introduction to computational fluid dynamics – the finite volume method*

The author of these slides wishes to thank these sources for making the current lecture.

Finite Volume Method in Multiple Dimensions – Convection-Diffusion

- Let's consider a convection-diffusion differential equation the describes the change of a conserved quantity,ϕ, w.r.t. time and space:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{u}) = \nabla \cdot (D \nabla \phi) + R \tag{1}$$

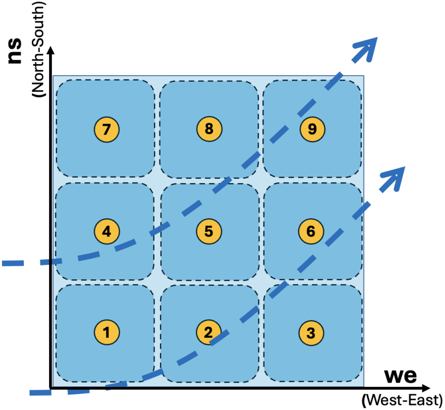
- Where:
  - ϕ is the quantity being conserved
  - $\frac{\partial \phi}{\partial t}$: is the rate of change of the conserved quantity w.r.t. time
  - $\nabla \cdot (\phi \vec{u})$: Is the convection/advection of the conserved quantity, and $\vec{u}$ is the velocity field.
  - $\nabla \cdot (D \nabla \phi)$: Is the diffusion of the conserved quantity (with D being the diffusion constant).
  - R is the source/sink of the conserved quantity.
- The time dependent convection-diffusion equation can be re-written as:

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (D \nabla \phi - \phi \vec{u}) + R \tag{2}$$

Control Volume Mathematics – Finite Volume Method

- Let's start by dividing the domain into a series of discrete volumes **over each of which** the following volume integral will be satisfied:

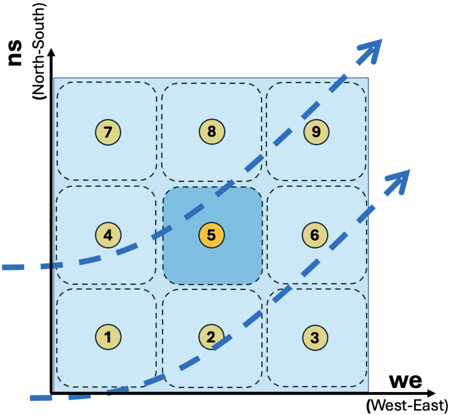
$$0 = \iiint_\Omega \left[\frac{\partial \phi}{\partial t} - \nabla \cdot \left(D\nabla\phi - \phi\vec{u} \right) - R \right] dV \tag{3}$$



Control Volume Mathematics – Finite Volume Method

- We can write the equation as individual integrals which would together apply to each of the finite volumes:

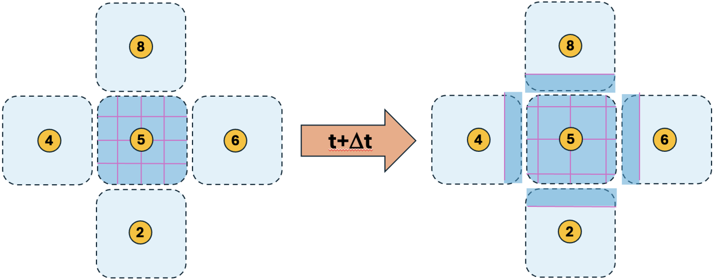
$$0 = \iiint_{\Omega 5} \frac{\partial \phi}{\partial t} dV + \iiint_{\Omega 5} \nabla \cdot (\phi \vec{u}) dV - \iiint_{\Omega 5} \nabla \cdot (D \nabla \phi) dV - \iiint_{\Omega 5} R dV$$



Control Volume Mathematics – Finite Volume Method

- Using the **divergence theorem** we can relate the divergence of the quantity to the flux of the quantity through the surface:

$$\iiint_{\Omega} \nabla \cdot (F) dV = \iint_{\Gamma} F \cdot \hat{n} dS \tag{4}$$



The **second** and **third** integrals in the expression can be written as:

$$0 = \iiint_{\Omega} \frac{\partial \phi}{\partial t} dV + \iint_{\Gamma} (\phi \vec{u}) \cdot \hat{n} dS - \iint_{\Gamma} (D \nabla \phi) \cdot \hat{n} dS - \iiint_{\Omega} R dV$$

Control Volume Mathematics – Finite Volume Method

- So, re-arranging the equation:

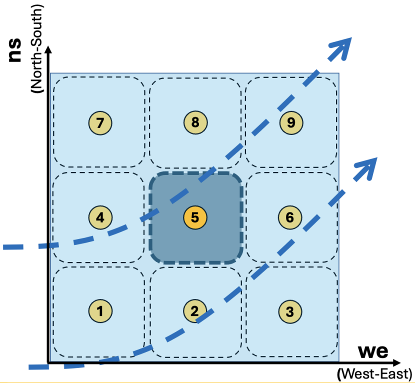
$$\underbrace{\iiint_\Omega \frac{\partial \phi}{\partial t} dV}_{Integral1} = \underbrace{-\iint_\Gamma (\phi \vec{u}) \cdot \hat{n} dS}_{Integral2} + \underbrace{\iint_\Gamma (D \nabla \phi) \cdot \hat{n} dS}_{Integral3} + \iiint_\Omega R dV$$

- Let's consider each integral one at a time. We'll dig into the details a bit.

Control Volume Mathematics – Finite Volume Method

- **Integral 1:** We will assume that ϕ represented as a function.
- **We will assume** a constant, cell-averaged value (not always the case).

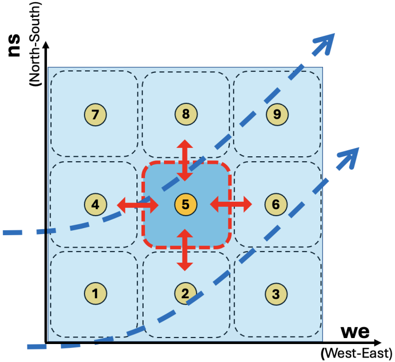
$$\iiint_\Omega \frac{\partial \phi}{\partial t} dV = \frac{\partial \bar{\phi}}{\partial t} \Delta x \Delta y$$



Control Volume Mathematics – Finite Volume Method

- **Integral 2:** Concerns the convective flux of ϕ across the cell boundaries. We will express this as a flux, F, across the boundary:

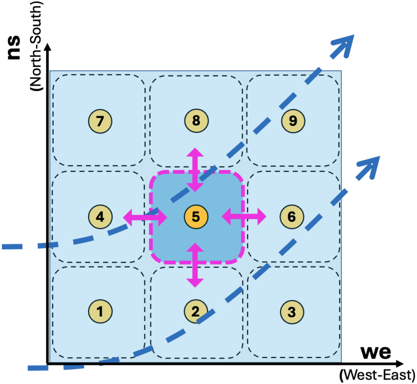
$$-\iint_{\Gamma} (\phi \vec{u}) \cdot \hat{n} dS = -(F_{east} - F_{west})\Delta y - (F_{north} - F_{south})\Delta x$$



Control Volume Mathematics – Finite Volume Method

- **Integral 3:** Concerns the diffusive flux of ϕ. We will express this as a flux, G, across the boundary:

$$\iint_{\Gamma} (D\nabla\phi) \cdot \hat{n} dS = (G_{east} - G_{west})\Delta y + (G_{north} - G_{south})\Delta x$$



Control Volume Mathematics – Finite Volume Method

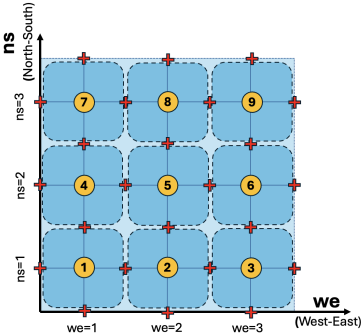
- At this point, we can combine the three expressions (and use abbreviations for faces):

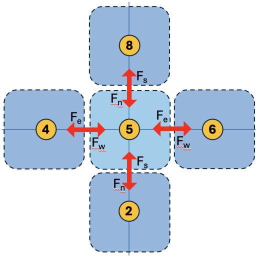
$$
\begin{aligned}
\frac{\partial \bar{\phi}}{\partial t}\Delta x \Delta y = & - && (F_E - F_W)\Delta y - (F_N - F_S)\Delta x \\
& + && (G_E - G_W)\Delta y + (G_N - G_S)\Delta x + \bar{R}\Delta x \Delta y
\end{aligned}
$$

Control Volume Mathematics – Finite Volume Method

- We will simplify by dividing both sides by $(\Delta x \Delta y)$:

$$\frac{\partial \bar{\phi}}{\partial t} = -\frac{1}{\Delta x}(F_E - F_W) - \frac{1}{\Delta y}(F_N - F_S) + \frac{1}{\Delta x}(G_E - G_W) + \frac{1}{\Delta y}(G_N - G_S) + \bar{R}$$





Control Volume Mathematics – Finite Volume Method

- Let's introduce a **simple** explicit **forward Euler/difference** formula for time integration (note we could have used an implicit or more accurate scheme here):

$$
\begin{aligned}
\frac{\phi_{i,j}^{k+1} - \phi_{i,j}^{k}}{\Delta t} = & - \frac{1}{\Delta x}(F_E - F_W) - \frac{1}{\Delta y}(F_N - F_S) \\
& + \frac{1}{\Delta x}(G_E - G_W) + \frac{1}{\Delta y}(G_N - G_S) + \bar{R}
\end{aligned}
$$

Let's Look at the discretization

- Rearranging and simplifying:

$$\begin{aligned}\phi_{i,j}^{k+1} &= \phi_{i,j}^{k} - \frac{\Delta t}{\Delta x}(F_E - F_W) - \frac{\Delta t}{\Delta y}(F_N - F_S) \\ &+ \frac{\Delta t}{\Delta x}(G_E - G_W) + \frac{\Delta t}{\Delta y}(G_N - G_S) + \bar{R}\Delta t\end{aligned}$$

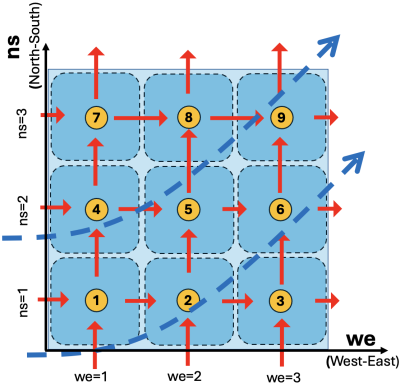
- Which is an equation to update ϕ (at the cell centers).
- In the following slides further assumptions are made to create a **simple FVM approach**.

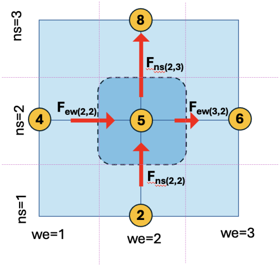
Let's Look at the discretization

- **Let's simplify:** consider only the **convective** fluxes
- The equation for updating $\phi_{i,j}^{k}$ in time is:

$$\phi_{i,j}^{k+1} = \phi_{i,j}^{k} - \frac{\Delta t}{\Delta x}(F_E - F_W) - \frac{\Delta t}{\Delta y}(F_N - F_S) + \bar{R}\Delta t$$

- The convective flux needs to capture the $F \simeq [\phi \vec{u} \cdot \hat{n}]$ across the faces:





Implementation of the Upwind Flux

- We need a numerical expression for the convective flux. Let's use the simplest one possible, the **upwind flux**:

$$F^x_{i-1/2,j} = F^x_{W,j} = \begin{cases} u_{i-1/2,j}\phi_{i-1,j}, & u_{i-1/2,j} > 0, \\ u_{i-1/2,j}\phi_{i,j}, & u_{i-1/2,j} \leq 0. \end{cases} \tag{5}$$

$$F^y_{i,j-1/2} = F^y_{i,S} = \begin{cases} v_{i,j-1/2}\phi_{i,j-1}, & v_{i,j-1/2} > 0, \\ v_{i,j-1/2}\phi_{i,j}, & v_{i,j-1/2} \leq 0. \end{cases} \tag{6}$$

- Note: We found it convenient (oncer we started coding) to define $F^y_{W,j}$ and $F^y_{i,S}$.
- To avoid using *if statements in the code*, this could be written more generally as:

$$F^x_{E,j} = \frac{1}{2}u_{i+1/2,j} \cdot (\phi_{i+1,j} + \phi_{i,j}) - \frac{1}{2}|u_{i+1/2,j}| \cdot (\phi_{i+1,j} - \phi_{i,j}) \tag{7}$$

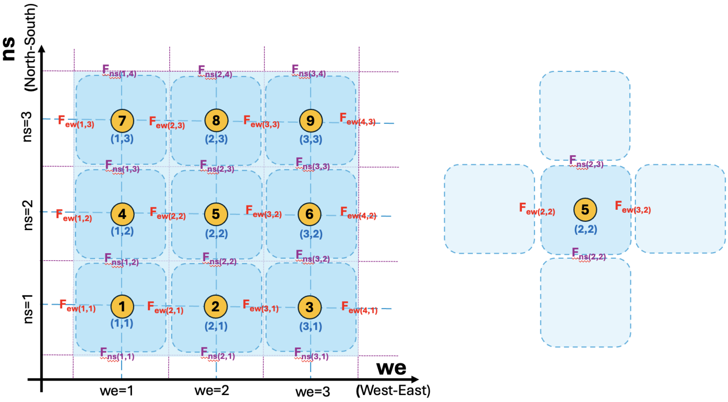
$$F^x_{i,N} = \frac{1}{2}u_{i,j+1/2} \cdot (\phi_{i,j+1} + \phi_{i,j}) - \frac{1}{2}|u_{i,j+1/2}| \cdot (\phi_{i,j+1} - \phi_{i,j}) \tag{8}$$

Finite Volume Method – Setting up the Finite Volume Method

- Let's implement this in code now!

MATLAB: Setting up the Finite Volume Method

- **Time & Geometry Discretization:** We'll define volumes (we use cell-centered, but other options exist).
- **Time discretization**: Timestep Δt will be important (we chose a simple explicit method, but other options exist)



MATLAB: Setting up the Finite Volume Method

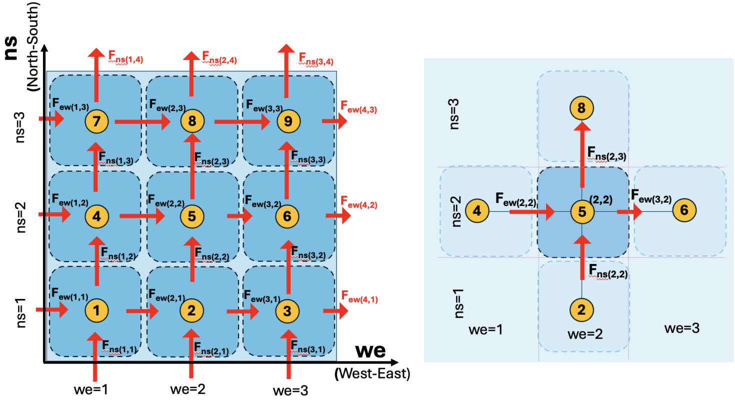
A few details about the implementation I will show (you or others may decide to approach the discretization and calculations differently).

- $N-1$: Number of cells in x-direction
- $M-1$: Number of cells in y-direction
- $\Delta x = 1/(N-1)$: Cell width in x
- $\Delta y = 1/(M-1)$: Cell width in y
- N, and M: Number of edges in x and y.
- Cell/Face Indexing and Numbering: We'll use a similar 'trick' as in FDM:

```
counter = 0;
for(we = 1:N-1)
for(ns = 1:M-1)
counter = counter + 1;
NCellNum(we, ns) = counter;
end
end
```

MATLAB: Setting up the Finite Volume Method

- Let's look at this:

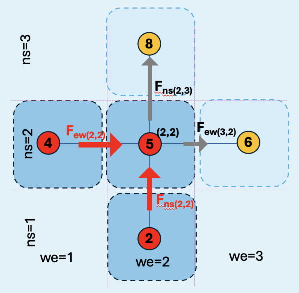


MATLAB: Setting up the Finite Volume Method

- Calculating the cell face fluxes using the **upwind convective flux***.
- **Note:** I have adjusted the flux eqns. to west and south faces based on my cell/face indexing!!!

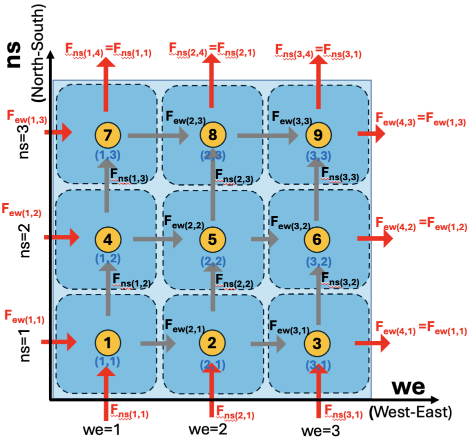
$$F^x_{i-1/2,j} = F^x_{WEST,j} = \begin{cases} u_{i-1/2,j}\phi_{i-1,j}, & u_{i-1/2,j} > 0, \\ u_{i-1/2,j}\phi_{i,j}, & u_{i-1/2,j} \leq 0. \end{cases} \tag{9}$$

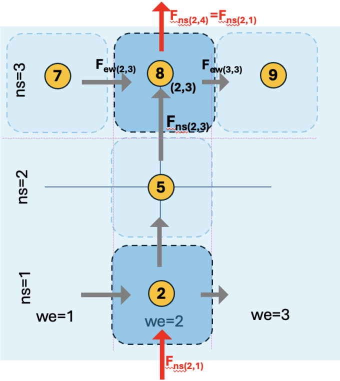
$$F^y_{i,j-1/2} = F^y_{i,SOUTH} = \begin{cases} v_{i,j-1/2}\phi_{i,j-1}, & v_{i,j-1/2} > 0, \\ v_{i,j-1/2}\phi_{i,j}, & v_{i,j-1/2} \leq 0. \end{cases} \tag{10}$$



Finite Volume Method – Setting up the Finite Volume Method

- Implementing periodic boundary conditions at the boundary cell faces using the **Upwind Convective Flux**:





Finite Volume Method – Setting up the Finite Volume Method

- Once we have the fluxes for all boundaries, we are able to update the cell-centered value.
- Explicit, forward Euler method for updating the conserved quantity in each cell:

$$\phi_{i,j}^{k+1} = \phi_{i,j}^{k} - \frac{\Delta t}{\Delta x}(F_E - F_W) - \frac{\Delta t}{\Delta y}(F_N - F_S) + \bar{R}\Delta t$$

Finite Volume Method – Setting up the Finite Volume Method

- Note: We added a small update in question 7 (implementing Lax-Wendroff flux) about direction/dimensional splitting – an approximate method to implement 1-D fluxes in 2D (this allows us to easily move into 2D on Cartesian grids).
- First we perform the time update in the x-direction (as a 1-D problem):

$$\phi^{*}_{i,j} = \phi^{k}_{i,j} - \frac{\Delta t}{\Delta x}(F_E - F_W) \tag{11}$$

- Then we calculate the fluxes in the y direction using the results of the step in x and complete the time-update:

$$\phi^{k+1}_{i,j} = \phi^{*}_{i,j} - \frac{\Delta t}{\Delta y}(F^{*}_N - F^{*}_S) \tag{12}$$

Finite Volume Method – What did we learn? and some conclusions/next steps

- We went over an introduction to FVM for 2-Dimensions.
- The upwinding method we show is conservative, but is numerically **diffusive** and **lacks accuracy**.
- We developed an FVM solver with periodic boundary conditions.
- There are (significantly) better approaches for the calculation of fluxes and the implementation of this problem.
- We saw despite the lack of accuracy/artificial diffusion, the FVM approach still conserved the quantity of interest.
- The CFD/transport communities have developed considerably better methods than this to solve complicated conservation problems – this provides an initial introduction. Take the CFD course!

Inclass Discussions On HW

- Today, we will spend 1/2 the class talking about HW 2, observations, and related ideas.
- Let's start with a quick MATLAB example of the HW problem, involving the convection (and diffusion) of ϕ
- Starting with the **upwind flux**
- Continuing with the **Lax-Wendroff**
- Then the concept of TVD/slope limiting fluxes.
- Commercial FVM solvers.

Inclass Discussions on FVM– Idea of Flux Limiters

- Let's consider the Lax-Wendroff flux from the HW:

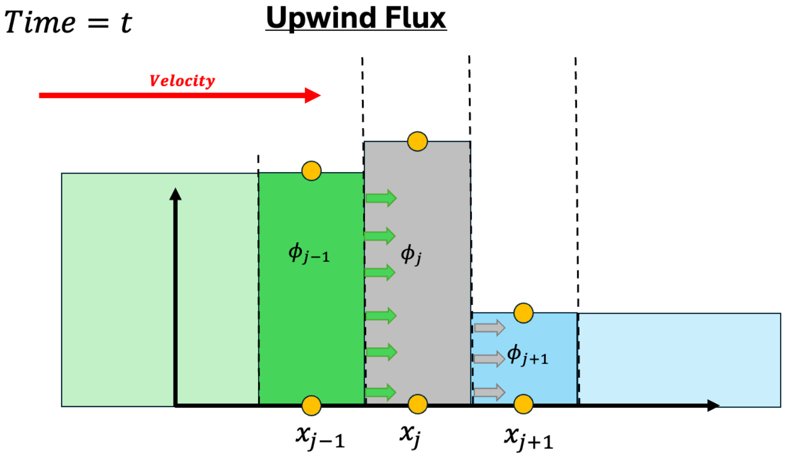
$$F^{LWx}_{j+1/2} = u_{j+1/2}\left(\frac{\phi_{j+1}+\phi_j}{2}\right) - u^2_{j+1/2}\frac{\Delta t}{2\Delta x}(\phi_{j+1}-\phi_j),$$

- And re-write it as:

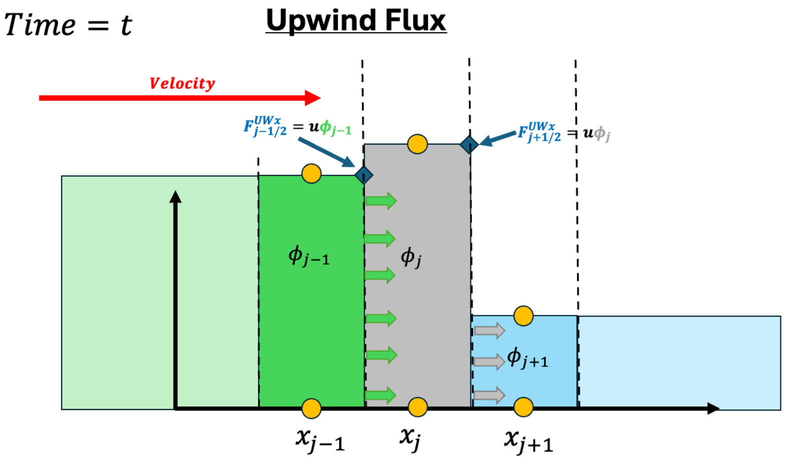
$$F^{LWx}_{j+1/2} = \underbrace{u\cdot\phi_j}_{upwindflux} + \underbrace{\frac{u}{2}\left(1-u\frac{\Delta t}{\Delta x}\right)\cdot(\phi_{j+1}-\phi_j)}_{LW-upwind:\ antidiffusive\ flux}$$

- The Anti-diffusive flux is the higher order term that reduces the (significant) diffusion from upwinding only.
- This anti-diffusive term is also responsible for the addition of oscillations to the solution!

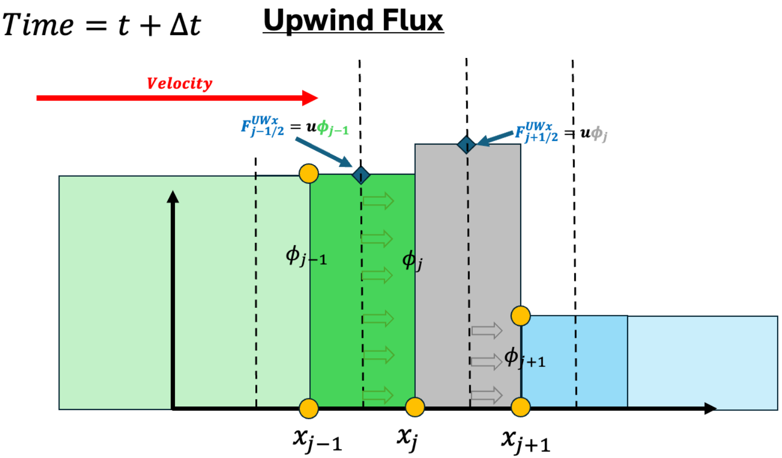
Examining Fluxes



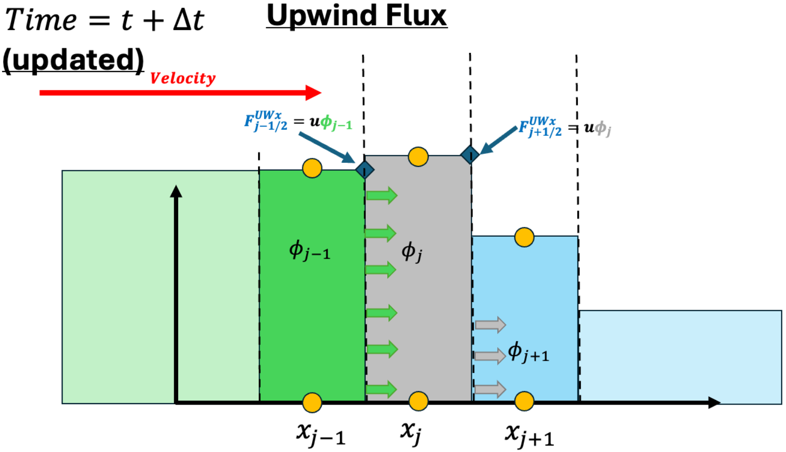
Examining Fluxes



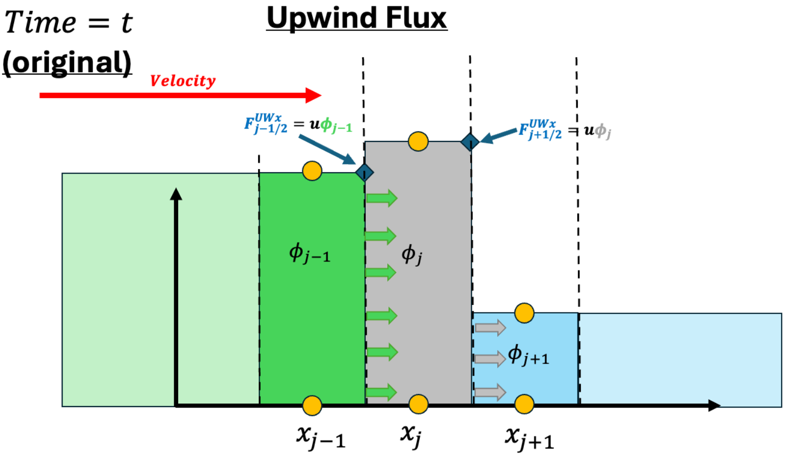
Examining Fluxes



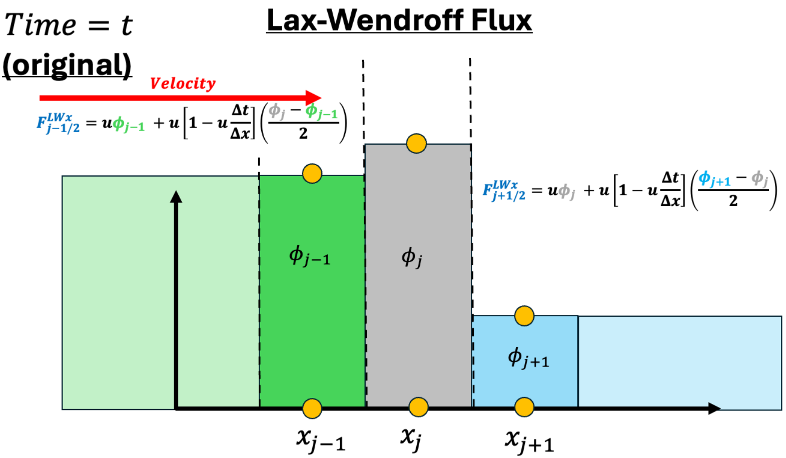
Examining Fluxes



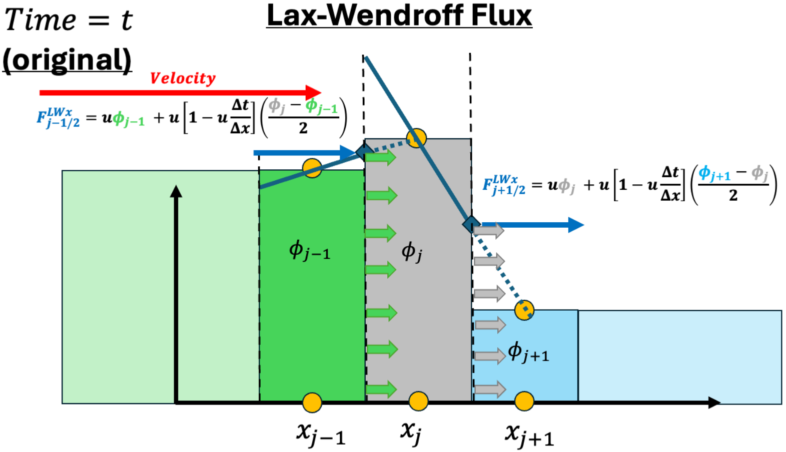
Examining Fluxes



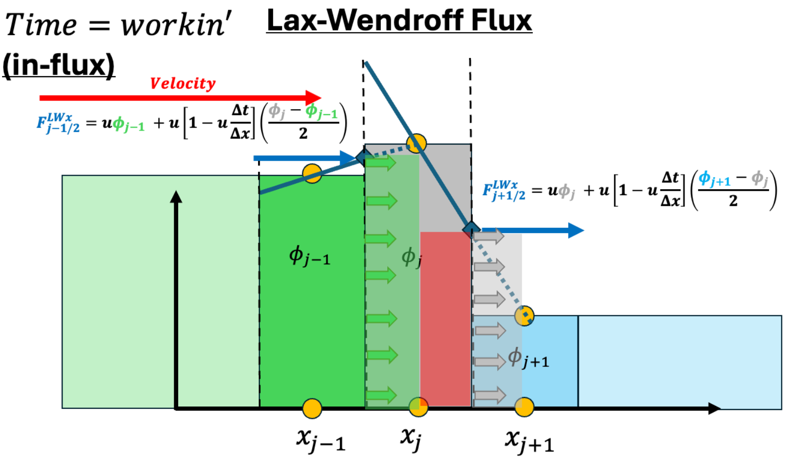
Examining Fluxes



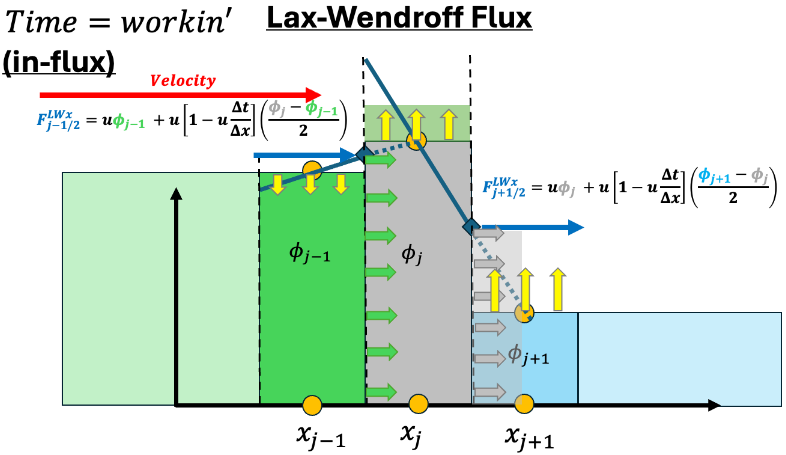
Examining Fluxes



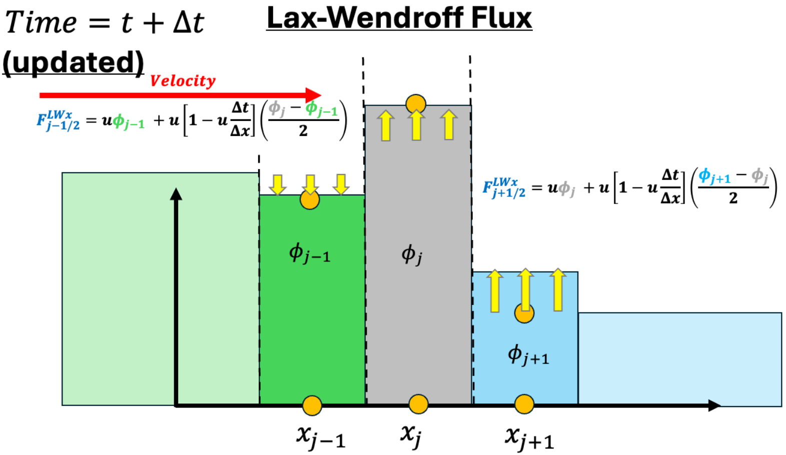
Examining Fluxes



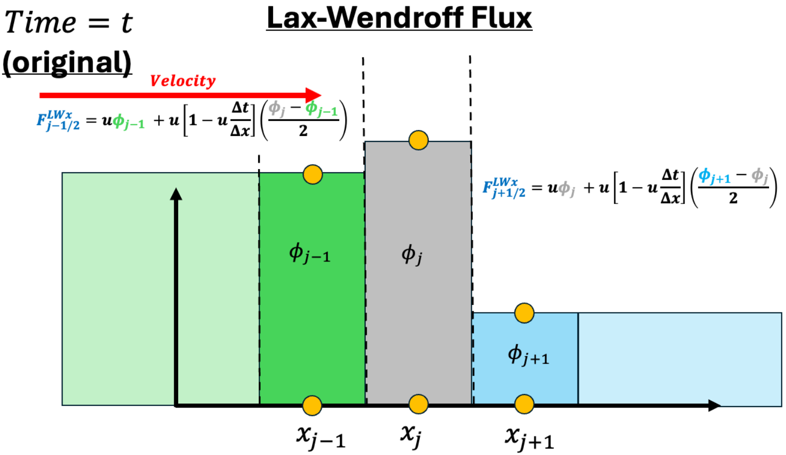
Examining Fluxes



Examining Fluxes



Examining Fluxes



Inclass Discussions on FVM– Idea of Flux Limiters

- Idea: Can we alter/adjust the amount of anti-diffusive flux? Yes (based on the criteria that the function be TVD)
- What can we use to determine when to alter the flux? Introduce a smoothness indicator:

$$r_j = \frac{\phi_j - \phi_{j-1}}{\phi_{j+1} - \phi_j} \tag{13}$$

- We can then introduce a flux limiter which adjusts the anti-diffusive term depending on the smoothness indicator, $\psi_{j+1/2}(r)$:

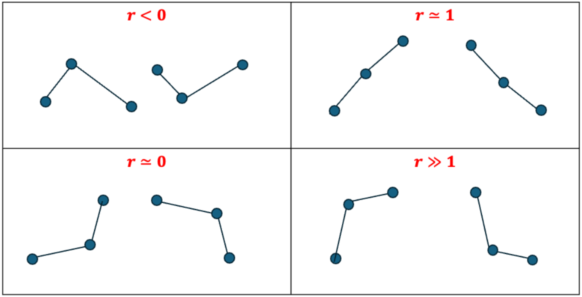
$$F^{LWx}_{j+1/2} = u \cdot \phi_j + \frac{u}{2}\left(1 - u\frac{\Delta t}{\Delta x}\right) \cdot \psi_{j+1/2} \cdot (\phi_{j+1} - \phi_j)$$

- What we really want to do is to "turn on" the limiter in regions where oscillations will appear.

Examining Fluxes

Let's check this out in MATLAB!!!!

$$r_j = \frac{\phi_j - \phi_{j-1}}{\phi_{j+1} - \phi_j} \tag{14}$$



Examining Fluxes

- Let's look at a few popular flux limiters:
- Recall, our smoothness sensor will be:

$$r_j = \frac{\phi_j - \phi_{j-1}}{\phi_{j+1} - \phi_j} \tag{15}$$

- Minmod Flux Limiter (most diffusive)

$$\psi_{MM}(r) = max([0, min([1, r])]) \tag{16}$$

- Van Leer Flux Limiter (happy medium):

$$\psi_{VL}(r) = \frac{r + |r|}{1 + |r|} \tag{17}$$

- Superbee Flux Limiter (least diffusive):

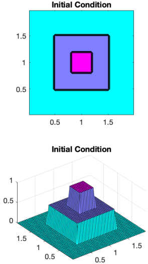
$$\psi_{SB}(r) = max([0, min([2r, 1]), min([r, 2])]) \tag{18}$$

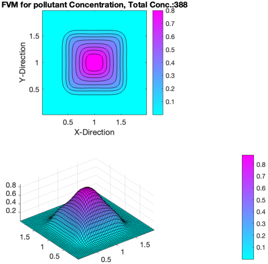
Examining Fluxes: Minmod

- Minmod Flux Limiter (most diffusive)

$$\psi_{MM}(r) = max([0, min([1, r])]) \tag{19}$$

- Tracks the "Lower Boundary" of the TVD region.



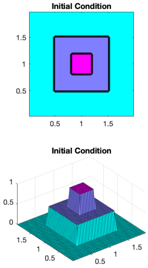


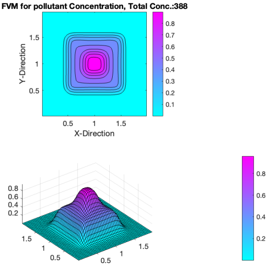
Examining Fluxes: Van Leer

- Van Leer Flux Limiter (happy medium):

$$\psi_{VL}(r) = \frac{r + |r|}{1 + |r|} \tag{20}$$

- Tracks "through" the TVD region (neither upper or lower boundary, but between them).



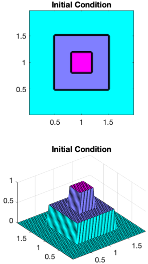


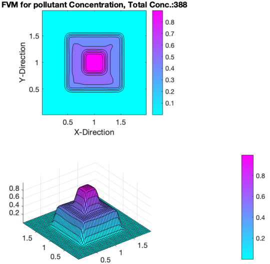
Examining Fluxes: Superbee

- Superbee Flux Limiter (least diffusive):

$$\psi_{SB}(r) = max([0, min([2r, 1]), min([r, 2])]) \tag{21}$$

- Is the upper boundary of the TVD region.





Inclass Discussions On HW

- Today, we will spend 1/2 the class talking about HW 2, observations, and related ideas.
- Let's start with a quick MATLAB example of the HW problem, involving the convection (and diffusion) of ϕ
- Starting with the **upwind flux**
- Continuing with the **Lax-Wendroff**
- Then the concept of TVD/slope limiting fluxes.
- Commercial FVM solvers.